



Commodity volatility shocks and BRIC sovereign risk: A GARCH-quantile approach

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ABSTRACT

We study whether the contemporaneous and lagged volatility of the commodity/energy markets can help predict the volatility of Brazil, Russia, India, China (BRIC) sovereign risk in the quantiles. We first define the latent volatility using Generalized Autoregressive Conditional Heteroscedasticity (GARCH) specifications and then use it within a quantile-based regression augmented with a dichotomous variable identifying the periods before and after the mid-2014 energy price decline. The analysis on daily data from January 4, 2010 to August 31, 2016 provides evidence that the volatility relation between commodity/energy and sovereign Credit Default Swap (CDS) markets is not the same under different volatility conditions, although energy volatility is slightly more important than commodity volatility. Importantly, the volatility of commodity/energy markets shapes sovereign risk in middle and upper volatility quantiles of commodities/energy exporters (Brazil and Russia). While in commodities/energy importers, such as China, predictability is only significant in upper volatility quantiles. Notably, we reveal the importance of the mid-2014 energy price decline for the volatility dynamics of BRIC sovereign risk; after the mid-2014, the volatility of the sovereign risk has increased in Brazil and Russia, while it has decreased in India. Finally, we recommend adoption of measures to reduce commodity and energy dependency.

1. Introduction

Unlike advanced economies (e.g. Norway, Australia and Canada), most of emerging economies are less diversified, have lower credit rating, and still depend on commodities and energy exportations (UNDP, 2011). Bhar and Nikolova (2009) argue that emerging economies are more vulnerable to energy price volatility than developed countries because they grow more rapidly and have more energy intensive production structures. Importantly, the price volatility of commodity and energy leads to macroeconomic instabilities as well as to a volatility in export earnings, foreign exchange reserves, current account balances, economic activities, and eventually to difficulties in meeting debt obligations. Hilscher and Nosbusch (2010) argue that a country with historically higher macroeconomic volatility is more prone to default.

However, the impact of volatility of (energy) commodities on the volatility of sovereign CDS spreads in large emerging countries remain extremely understudied, although numerous press articles have focused

on the negative impact of the sharp energy prices decline since mid-2014 on the sovereign risk of emerging energy-exporting countries.¹ For example, lower energy prices deteriorated fiscal balances in Russia that is strained to fund its public budget and pay off its external dollar-denominated debt. Sharma and Thuraisamy (2013) highlight the role of oil price uncertainty in predicting sovereign CDS returns in eight Asian countries but don't account for the effect of this role on the volatility of CDS spreads. Liu et al. (2016) focus on the statistical properties of country risk rating in oil-exporting countries and show that oil price volatility can accentuate the volatility of country risk ratings. Lee et al. (2017) show that country risk in net oil-importing (Germany, France, Italy, Japan and US) and net oil-exporting countries (Canada, UK) is affected by oil price shocks. An interesting study by Bouri et al. (2017) focuses on the volatility spillover between commodities and sovereign CDS spreads of emerging and frontier economies and reports significant volatility linkages that were found not be closely related to the levels of commodities and energy dependence. However, Bouri et al. (2017) just concentrate on the average dependence in the volatility without taking

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¹ <http://www.markit.com/Commentary/Get/04032015-Credit-Brazil-s-CDS-spreads-jump-above-Latam-peers>.
<http://www.reuters.com/article/russia-cds-idUSL6N0VC2E220150202>.

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into account the difference in the dependence between high and low quantiles. It is documented that the return and volatility distribution of assets is most often heterogeneous, suggesting the appropriateness of using a quantile approach to uncover differences in the response of the volatility of sovereign CDS spreads to changes in the volatility of commodities across the different quantiles.

We therefore employ a quantile-based approach to uncover the impact of quantile (energy) commodity price volatility on different volatility of sovereign CDS spreads² quantiles, accounting for the effect of the mid-2014 energy price decline.³ [Reboredo and Uddin \(2016\)](#) have recently employed a quantile regression to account for the impact of financial stress and policy uncertainty on changes in commodity futures. They argue that the quantile regression allows examination of co-movement and Granger causality in specific market conditions (e.g. low, median, and high quantiles) that other models (e.g., switching models) are incapable to capture.⁴ This is important given our analysis involves volatility series modeled via a Generalized Autoregressive Conditional Heteroscedasticity (GARCH) framework.⁵ Low, median, and high quantiles correspond respectively to low volatility, moderate volatility, and high volatility conditions. Measuring how co-movement and Granger causality between the volatility of commodity/energy prices and the volatility of sovereign CDS spreads have changed in different volatility conditions has implications for policy-makers and investors in terms of risk management and the predictability of sovereign CDS spreads.

We consider Brazil, Russia, India, China (BRIC) countries, which represent a suitable and heterogeneous group of countries to conduct our analysis given they include major energy exporters and energy exporters. BRIC countries also attract a large portion of capital inflows and represent a major investment destination for global portfolio managers. Often the decision of global investors to get into or out of emerging economies depends among others on sovereign risk, as measured by the sovereign CDS spread. The latter is a risk measure closely watched by risk managers and policy-makers as it represents the cost at which a country can raise fund in international markets. For example, higher sovereign CDS spreads suggest that a country may face constraints in raising funds abroad and in attracting foreign investments.

Our main results show that the dynamics of the volatility of energy commodities have an impact on the volatility of sovereign CDS spreads, and this impact differs by country and between lower, intermediate and upper quantiles. Importantly, we reveal that, in most cases, the mid-2014 energy price decline has increased the volatility of sovereign risk in Brazil and Russia and decreased it in India.

The layout of the paper is as follows. [Section 2](#) reviews the related literature. [Section 3](#) describes the empirical model. [Section 4](#) provides the data and its preliminary analysis. [Section 5](#) presents and discusses the empirical results. [Section 6](#) summarizes our concluding remarks.

2. Related works

The effect of energy commodities on stock market returns has been extensively studied in both developed and emerging countries (see, among others, [Cuando and de Garcia, 2014](#); for developed countries;

and [Basher et al., 2012](#); [Ghosh and Kanjilal, 2014](#); [Bouri, 2015](#) for developing and emerging countries). However, limited studies have so far considered that effect in the sovereign CDS markets.

Along with globalization and the rising importance of the sovereign CDS in mitigating credit risk, numerous studies emerge to understand the determinants and the interaction of sovereign CDS spreads with other economic and financial variables, especially in emerging countries that attract important international capital. Some studies focus on the role played by commodities in determining sovereign CDS spreads. In a comprehensive study, [Longstaff et al. \(2011\)](#) use monthly data from 26 emerging economies and decompose the determinants of sovereign Credit Default Swap (CDS) into country-specific and global factors. They stress on the importance of global factors as a major drivers of CDS spreads and highlight the role of the commodity term of trade. [Fonseca et al. \(2016\)](#) stress on the importance of volatility and jumps extracted from the oil futures in explaining CDS spread changes in the energy sector. [Aizenman et al. \(2016\)](#) concentrate on the sovereign risk of 18 emerging and 10 developed countries and show that the commodity term of trade volatility is positively associated with sovereign CDS spreads. [Hilscher and Nosbusch \(2010\)](#) examine the effects of macroeconomic fundamentals on sovereign credit spreads in 31 emerging markets from 1994 to 2007 and show that the countries that are more susceptible to external shocks are those with larger fluctuations in their terms of trade, possibly due to concentrated exports in commodities with volatile prices. [Sun et al. \(2011\)](#) focus on the determinants of emerging markets sovereign spreads accounting for price changes in commodity prices. The authors show that commodity price increases are associated with lower sovereign spreads. [Arezki and Bruckner \(2010\)](#) stress on the importance of changes in international commodity prices on the sovereign spreads in emerging markets through their effect on foreign currency revenues. Though their results show that positive commodity price shocks generally reduce the sovereign bond spread, this reduction depends also on the level of democracy. Few other studies include the impact of oil prices while modelling sovereign spreads. [Duffie et al. \(2003\)](#) show that the Russian credit risk responds to political events and is negatively correlated with Russian foreign currency reserves and the oil price. [Alexandre and de Benoist \(2010\)](#) examine the impact of oil price returns and volatility on bond risk premiums in seventeen emerging countries from 1998 to 2008. The authors stress on the important role played by oil price return and volatility in determining sovereign credit risk, especially in oil-exporting countries (i.e. Russia and Brazil). [Sharma and Thuraissamy \(2013\)](#) focus on eight Asian countries and find that oil price uncertainty can be used as a predictor for CDS returns in several cases. Instead, we concentrate on the effect of both commodities and energy commodities on the volatility of CDS spreads. [Hooper \(2015\)](#) examines the association between oil and gas reserves and sovereign spreads in 10 emerging oil-exporting countries from 1994 to 2014. The author shows that oil reserves have an effect on sovereign spreads but this effect also depends on the institutional quality of the country, namely corruption, political stability, and democracy. [Liu et al. \(2016\)](#) show that oil price volatility can accentuate the volatility of country risk ratings. [Lee et al. \(2017\)](#) using a structural VAR approach to monthly data and show that country risk in net oil-importing (Germany, France, Italy, Japan and US) and net oil-exporting countries (Canada, UK) is affected by oil price shocks. We instead focus on the volatility of sovereign risk, as measured by CDS spreads, using a daily dataset. We also account for the impact of the mid-2014 energy price decline. In an interesting paper, [Bouri et al. \(2017\)](#) examine explicitly the volatility transmission from commodity and energy indices to sovereign CDS spreads in emerging and frontier economies. Based on a causality in variance test, the authors reveal that the volatility transmission from commodities to CDS spreads is not a clear cut as it doesn't depend solely on commodity and energy dependence. However, [Bouri et al. \(2017\)](#) didn't differentiate between upper and lower tail dependence in the dynamics of the causality in variance. We contribute to the finance literature by uncovering the volatilities

² [Bouri et al. \(2017\)](#) define sovereign CDS as “bilateral contracts between a buyer and a seller where the seller is offering protection against credit event by a sovereign borrower. The buyer pays a premium to the protection seller in exchange for compensation in case of a credit event. The CDS premium is quoted as a fraction of the notional value of the reference obligation (in basis points). The failure of a sovereign borrower to meet debt obligations is known as a credit event. Qualifying credit events include failure of the sovereign borrower to pay principal or interest payments, restructuring or moratorium”.

³ Although various economic and political factors may also affect the volatility of sovereign risk, we focus in this study on the volatility linkages from energy and commodity markets to sovereign risk.

⁴ [Reboredo and Ugolini \(2017\)](#) also employ a quantile-based approach to study the dependence between gold commodity and gold stock returns.

⁵ We define the latent volatility using GARCH specifications.

interdependence between (energy) commodities and sovereign CDS spreads of BRIC countries under different market conditions (i.e. in lower, median, and upper quantiles), considering for the impact of the mid-2014 energy price decline.

3. Research method

We employ a GARCH-quantile regression within a two-step approach. In the first, we fit the best mean and variance GARCH equation to the return series. In the second, we use the conditional volatility (GARCH) series extracted from the first step within a quantile regression approach in order to capture the contemporaneous and lagged dependence structure among the volatilities of (energy) commodities and sovereign CDS spreads. In doing so, we account for the impact of the mid-2014 energy price decline on the volatility of CDS spreads by using a dummy variable.

3.1. GARCH conditional volatility

GARCH-based models are powerful tools to model the volatility of financial variables and to capture several of their stylized features such as volatility clustering, heteroscedasticity, and non-normality, and asymmetry in the response of volatility to positive and negative shocks. Unlike the standard GARCH model of Bollerslev (1986), the asymmetric GARCH of Glosten et al. (1993), popularly known as GJR-GARCH model, adds an asymmetric term to capture the asymmetric response of the conditional variance to shocks. The standard GARCH and the asymmetric GJR-GARCH model are given respectively in Eq. (1) and Eq. (2), respectively.

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (1)$$

where σ_t^2 is the conditional variance, ε_{t-1} is the innovation, α represents the ARCH term which measures the impact of past innovations on current variance, and β represents the GARCH term which measures the impact of past variance on current variance.

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 + \gamma \varepsilon_{t-1}^2 I_{\varepsilon < 0} \quad (\varepsilon_{t-1}) \quad (2)$$

where I is a dummy variable that measures the asymmetric response of the conditional variance to shocks. It takes a value of unity in response to negative shocks and zero in response to positive shocks. In addition to the constraints required by the standard GARCH to ensure stationarity and positivity ($\omega > 0$; $\alpha \geq 0$; $\beta \geq 0$; and $\alpha + \beta < 1$), an additional constraint has to be respected in the case of the GJR-GARCH ($\alpha + \beta + 0.5\gamma < 1$).

To decide which GARCH-based model has a superior fit (the order of the AR specification in the mean equation⁶; the type of the GARCH specification, i.e. symmetric (GARCH) or asymmetric (GJR-GARCH); and the density of the error distribution, normal, t-student, or generalized error distributions), we follow Beine and Laurent (2003) and use SIC. We also conduct several diagnostic tests for the residuals and the squared residuals to evaluate the goodness of fit of the selected models. Compared to an ad-hoc selection, such a well-developed pre-selection in the univariate GARCH modelling would potentially reinforce the efficiency of the related estimators and thus lead to more valid estimates of the volatility relationships, as captured by the quantile regression approach.

3.2. Quantile regression

The quantile regression has proven to be a very precise tool to model the impact of the conditional variables on the dependent variable (Koenker, 2005). Introduced by Koenker and Bassett (1978), the

quantile regression approach can help to assess whether there is a relation at the conditional median, or other conditional quantiles. As such, the quantile regression provides a nuanced representation of the stochastic interdependence among random variables, which is very informative especially when the conditional distribution is heterogeneous. This is crucially important for the purpose of our study which is to identify the relation between the volatility of commodity/energy and that of sovereign CDS spreads under different market conditions. While the copula function can provide the dependence measure for the lower and the upper tail, as well as the average dependence, the quantile regression differs in that it directly relates the quantile of the dependent variable with the conditioning variables; whereas the copula function directly relates the quantiles of both the conditioning variables. Numerous recent studies have applied the quantile regression approach (e.g. Reboredo and Uddin, 2016).

We consider that the conditional τ quantile of the volatility of CDS spreads distribution is affected by the contemporaneous and lagged volatility of the commodity and energy markets, while accounting for the impact of the mid-2014 energy price decline.

$$Q_{VCDSt}(\tau | VCDSt_{t-1}, d_t, X_t, X_{t-1}) = \alpha(\tau) + \beta(\tau)VCDSt_{t-1} + \delta(\tau)d_t + \phi(\tau)X_t + \varphi(\tau)X_{t-1} \quad (3)$$

where $VCDSt_t$ represents the volatility of CDS spreads of a country; $VCDSt_{t-1}$ is the lagged variable of the dependent variable; d_t is the dummy variable identifying the period before and after the mid-2014 energy decline denoted as 0 and 1, respectively; X_t represents a $k \times 1$ vector of regressors, which is the constant and the energy (commodity) price index.

For a given τ , the parameters in Eq. (3) are estimated by minimizing the weighted absolute deviation as follows:

$$\arg \min_{\alpha(\tau), \beta(\tau), \delta(\tau), \phi(\tau), \varphi(\tau)} \sum_{t=1}^T \rho_{\tau}(VCDSt_t - \alpha(\tau) - \beta(\tau)VCDSt_{t-1} - \delta(\tau)d_t - \phi(\tau)X_t - \varphi(\tau)X_{t-1}) \quad (4)$$

where $\rho_{\tau}(u) = u(\tau - I(u < 0))$, $0 < \tau < 1$, and $I(\cdot)$ represents the indication function.

The minimization problem in Eq. (4) is solved using the linear programming methods of Koenker and d'Orey (1987). The pair bootstrap method of Buchinsky (1995) is used to obtain the standards errors for the estimated coefficients. This method is used because it provides asymptotically valid standard errors under misspecifications of the quantile regression function and under heteroscedasticity.

4. Data and preliminary analysis

This study is conducted with daily sovereign CDS spread changes for BRIC countries (Brazil, Russia, India and China) and logarithmic returns for the commodity and energy indices (the Standard and Poor's Goldman Sachs Commodity Index (S&P GSCI) and the S&P GSCI energy sub-index). Specifically, the CDS spread for the 5-year maturity contract is used as it represents the most traded and liquid contract. The S&P GSCI aggregate index and the energy sub-index represent the most widely recognized benchmark for the market of global commodity and energy commodity, respectively. The aggregate S&P GSCI contains 25 commodities, seven of which are energy commodities. In the aggregate S&P GSCI index, the weight of energy commodities (crude oil, natural gas, heating oil, and RBOB gasoline) is 63.05% whereas the weight of non-energy commodities (agriculture, livestock, industrial metals, and precious metals) is 36.95%. Data are collected from DataStream for the period from January 4, 2010 to August 31, 2016. Notably, the sample period starts 6 months after the end of the US recession, as defined by the National Bureau of Economic Research (NBER).

⁶ Notably, the mean equation of the GARCH-based model is specified as an autoregressive model.

Table 1
Statistical properties of daily data.

	Brazil	Russia	India	China	Commodity	Energy
Mean	0.069	0.010	0.057	0.014	0.000	0.000
Maximum	38.46	110.15	45.85	24.93	0.055	0.088
Minimum	−83.957	−85.62	−58.21	−19.553	−0.067	−0.093
Std. Dev.	6.888	13.074	4.966	3.298	0.012	0.017
Skewness	−0.755	0.135	−0.613	0.508	−0.192	−0.030
Kurtosis	19.759	15.548	30.484	11.166	5.3.000	6.134
Jarque-Bera	20495.3***	11414.36***	54840.85***	4906.285***	393.996***	712.08***
ADF	−21.599***	−57.339***	−25.260***	−25.029***	−42.431***	−43.175***
PP	−39.268***	−59.762***	−41.554***	−39.999***	−42.428***	−43.153***
ARCH(10)	32.642***	74.379***	6.895***	32.253***	10.805***	19.835***

Notes: This table presents the statistical properties of 1738 daily observations for the period January 4, 2010 – August 31, 2016. Both unit root tests (ADF and PP) are conducted with an intercept; ARCH represents statistics for Engle's heteroscedasticity test.

*** $p < 0.01$.

Table 2
Commodity Exports per Country.
Source: UNCTAD (2014) http://unctad.org/en/PublicationsLibrary/suc2014d7_en.pdf.
Russian data are obtained from <http://www.imf.org/external/pubs/ft/weo/2015/02/>.

	Commodity exports as % of merchandise exports	Commodity exports as % of GDP	Exports by commodity group	
			Energy	Non- energy
BRIC countries				
Brazil	65	6.8	14	86
Russia	60.5	8.1	83.1	16.9
India	47	7.6	43	57
China	8	2.5	23	77

The statistical properties for the 5-year CDS spread changes and the logarithmic returns for the commodity and energy indices are given in Table 1. Commodity and energy returns have lower standard deviation than the CDS spreads, whereas Brazil and Russia have the highest standard deviation of CDS spread changes among the BRIC countries. All data series exhibit excess kurtosis and non-zero skewness, confirming their departure from normality as shown by the Jarque-Bera normality test. Given that the GARCH and regression estimations require stationary series to eliminate spurious problems, the presence of a unit root test is verified with augmented Dicky-Fuller and Phillips-Perron tests. The results show that all the return and CDS spread changes are stationary. The ARCH test indicates that heteroscedasticity is present, suggesting the usefulness of employing GARCH-based models.

For each of BRIC countries, Table 2 presents the main commodity-related exports as a percent of GDP and as a percent of exports.⁷ Commodity exports account for at least 60% of merchandise exports in Russia and Brazil. They also account for more than 6% of GDP in India and Brazil. In examining the exports by commodity group, energy commodities are over 83% of Russia's commodity exports, followed by 43% for India. Non-energy commodities are the majority of exports in Brazil and China with 86% and 77% of commodity exports, respectively. It is worth noting here that China and India are among the largest consumer and importer of crude oil. In fact, crude petroleum imports represent more than 33% of India's total imports and around 14% of China's total imports. The EIA indicates that China and India are net oil importers.⁸ This suggests the sensitivity of the Indian and Chinese economies and potentially their CDS spreads to the condition and price volatility in the global energy market.

5. Empirical results

5.1. GARCH volatility series

In modelling the conditional volatility of each of the series, a maximum of two lags in the specification of the mean equation was the optimal choice to control for historical patterns in commodity/energy returns and CDS spread changes, whereas the GARCH (1,1) outperformed the GJR-GARCH model in all cases. As shown in Table 3, the ARCH and GARCH parameters are positive and significant at the 1% significance level in most cases. More importantly, the stationarity and positivity constraints have been respected, and the diagnostic results show that heteroscedasticity is not present in the residual series. Given that the main purpose of GARCH modelling is just to extract the conditional volatility series and used them in the quantile regression, we do not elaborate on the results of the estimated GARCH model coefficients.

5.2. GARCH-quantile regression

Having extracted the GARCH volatility series, the next step consisted of using them within a quantile regression to estimate the contemporaneous and lagged volatility transmission between commodity/energy prices and CDS spread changes of BRIC countries. We also account for the impact of the energy price decline of mid-2014 by adding a dummy variable.⁹ The coefficients estimates for the 5th, 10th, 30th, 50th, 70th, 90th, and 95th quantile regression for Brazil, Russia, Indian and China are given respectively in Tables 4–7.

The results from Table 4 show that, in Brazil, the contemporaneous effect of commodity volatility on the volatility of CDS spreads is positive at the 30th quantile and above, and exhibits stronger relation as we move higher in the quantiles. Concerning causality, past values of commodity volatility have a negative impact at the 30th quantile and above. These results confirm the relevance of commodity volatility in shaping the volatility of CDS spreads in Brazil, and are in line with the high percentage of commodity exports as % of merchandise exports shown in Table 2. As for the coefficient of the dummy variable, it is significantly positive at the 70th quantile, suggesting that the decline in the price of (energy) commodities since mid-2014 has increased the volatility of the Brazilian sovereign risk at quantile 0.70. This latter result suggests that investors and policy-makers have to worry about the effect of declining energy commodities prices on the volatility of Brazilian CDS spreads at a the 70th quantile only. As expected, the lagged values of the volatility of the CDS spread are significantly positive across all the quantiles and the impact is stronger in higher

⁷ Source http://unctad.org/en/PublicationsLibrary/suc2014d7_en.pdf.

⁸ <https://www.eia.gov/tools/faqs/faq.php?id=709&t=6>.

⁹ It is worth noting here that, given the large difference in the scale of the GARCH volatility series between CDS and (energy) commodities, we standardized all the GARCH volatility series before using them in the quantile regression.

Table 3
Estimation of the selected GARCH-based models.

	Brazil	Russia	India	China	Commodity	Energy
<i>Mean equation</i>						
Constant	−0.048	−0.137	−0.005	−0.001	0.000	0.000
AR(1)	0.102***	–	0.033	–	–	–
AR(2)	–	–	0.085***	–	–	–
<i>Variance equation</i>						
Constant	0.255***	0.983***	0.032***	0.256***	0.000 ⁺	0.000 ⁺
ARCH	0.143***	0.183***	0.070***	0.135***	0.045***	0.049***
GARCH	0.860***	0.809***	0.883***	0.846***	0.953***	0.950***
GED parameter	–	–	–	–	1.310***	1.258***
Student (DoF)	–	7	8	–	–	–
<i>Model diagnostics</i>						
ARCH(10)	1.286	0.781	0.349	0.642	0.878	0.748

Notes: AR(1) and AR(2) denote respectively the value of the one- and two-lagged of the dependent variable; lagged variables; Student (DoF) indicates the student-t distribution parameter; ARCH (10) represents statistic tests of the null hypothesis of no conditional heteroscedasticity up to 10 lags for the residuals; the Box–Pierce Q-squared statistic tests the null hypothesis of no autocorrelation up to order 10 for squared values.

* $p < 0.10$.

*** $p < 0.01$.

Table 4
Estimation results for the quantile regression-Brazil.

<i>Dependent variable: Brazil</i>							
	Q(0.05)	Q(0.10)	Q(0.30)	Q(0.50)	Q(0.70)	Q(0.90)	Q(0.95)
<i>Independent variables</i>							
Commodity	0.000	0.001	0.018	0.055	0.168	0.313	0.441
Commodity(−1)	0.000	−0.001	−0.017	−0.053	−0.163	−0.301	−0.426
Dummy	0.000	0.001	0.002	0.004	0.017	0.043	0.049
Lagged	0.860	0.860	0.869	0.903	0.956	1.131	1.322
Constant	0.002	0.003	0.002	0.001	0.002	0.003	0.004
<i>Independent variables</i>							
Energy	0.000	0.001	0.021	0.077	0.212	0.285	0.515
Energy(−1)	0.000	−0.001	−0.020	−0.074	−0.204	−0.267	−0.500
Dummy	0.000	0.001	0.002	0.003	0.016	0.045	0.041
Lagged	0.860	0.860	0.869	0.903	0.955	1.131	1.327
Constant	0.002	0.002	0.002	0.001	0.002	0.003	0.007

Notes: This table provides the quantile regression estimates in Eq. (3) for the volatility of the BRIC country sovereign CDS spreads. Commodity, Energy, dummy, and lagged denote the commodity volatility, energy volatility, the mid-2014 decline in energy dummy variable, and the one-lagged value of the dependent variable. Daily data are from January 4, 2010 to August 31, 2016. Bold denotes significance at the 5% level.

Table 5
Estimation results for the quantile regression-Russia.

<i>Dependent variable: Russia</i>							
	Q(0.05)	Q(0.10)	Q(0.30)	Q(0.50)	Q(0.70)	Q(0.90)	Q(0.95)
<i>Independent variables</i>							
Commodity	0.000	0.001	0.014	0.049	0.082	0.169	0.286
Commodity(−1)	0.000	−0.001	−0.013	−0.047	−0.080	−0.165	−0.279
Dummy	0.000	0.000	0.001	0.003	−0.002	0.038	0.138
Lagged	0.809	0.810	0.814	0.834	0.987	1.249	1.338
Constant	0.003	0.003	0.002	0.003	0.001	0.003	0.015
<i>Independent variables</i>							
Energy	0.000	0.001	0.022	0.056	0.088	0.216	0.297
Energy(−1)	0.000	−0.001	−0.021	−0.053	−0.087	−0.215	−0.298
Dummy	0.000	0.000	0.000	0.003	−0.001	0.040	0.151
Lagged	0.809	0.810	0.814	0.834	0.985	1.248	1.332
Constant	0.003	0.003	0.002	0.003	0.001	0.007	0.024

Note: see notes to Table 4.

quantiles. Quite similar, but stronger, results are reported for the effects of energy volatility.

In Russia (see Table 5), the contemporaneous effect of commodity volatility is positive at the 30th quantile and above, whereas the lagged effect of commodity volatility is negative at the same quantiles. A one-unit increase in the commodity volatility variable is associated with an increase in the volatility of Brazilian CDS spreads by 0.014 points at the

30th quantile, 0.049 points at the median and 0.286 at the 0.95th quantile. The coefficient of dummy variable shows that the decline in the price of (energy) commodities since mid-2014 has increased the volatility of sovereign risk at the upper quantile of 0.95, suggesting that investors and policy-makers have to worry about the effect of declining energy commodities prices on the volatility of Russian CDS spreads at the extreme upper quantile of 0.95. The effect of lagged values of the

Table 6
Estimation results for the quantile regression-India.

<i>Dependent variable: India</i>							
	Q(0.05)	Q(0.10)	Q(0.30)	Q(0.50)	Q(0.70)	Q(0.90)	Q(0.95)
<i>Independent variables</i>							
Commodity	0.000	0.000	0.000	0.000	0.000	0.024	0.061
Commodity(−1)	0.000	0.000	0.000	0.000	0.001	−0.023	−0.047
Dummy	0.000	0.000	0.000	0.000	−0.008	−0.052	−0.179
Lagged	0.883	0.883	0.883	0.885	0.901	1.023	1.168
Constant	0.001	0.001	0.001	0.001	0.008	0.057	0.171
<i>Independent variables</i>							
Energy	0.000	0.000	0.000	0.000	0.000	0.003	−0.006
Energy(−1)	0.000	0.000	0.000	0.000	0.001	−0.003	0.015
Dummy	0.000	0.000	0.000	0.000	−0.008	−0.052	−0.190
Lagged	0.883	0.883	0.883	0.885	0.901	1.022	1.168
Constant	0.001	0.001	0.001	0.001	0.009	0.058	0.187

Note: see notes to Table 4.

Table 7
Estimation results for the quantile regression-China.

<i>Dependent variable: China</i>							
	Q(0.05)	Q(0.10)	Q(0.30)	Q(0.50)	Q(0.70)	Q(0.90)	Q(0.95)
<i>Independent variables</i>							
Commodity	0.000	0.000	0.000	0.012	0.036	0.294	0.446
Commodity(−1)	0.000	0.000	0.000	−0.011	−0.034	−0.278	−0.414
Dummy	0.000	0.000	0.000	−0.001	−0.003	−0.018	−0.058
Lagged	0.846	0.846	0.859	0.891	0.955	1.147	1.333
Constant	0.017	0.016	0.014	0.010	0.010	0.017	0.032
<i>Independent variables</i>							
Energy	0.000	0.000	0.000	0.019	0.033	0.239	0.612
Energy(−1)	0.000	0.000	0.001	−0.017	−0.030	−0.228	−0.583
Dummy	0.000	0.000	0.000	−0.002	−0.005	−0.022	−0.041
Lagged	0.846	0.846	0.859	0.890	0.955	1.142	1.341
Constant	0.017	0.016	0.014	0.010	0.011	0.033	0.042

Note: see notes to Table 4.

volatility of the CDS spread is significantly positive across all the quantiles and exhibits an upward trend. These results support the documented view on the importance of energy price volatility in affecting the volatility of Russian sovereign risk (Bouri et al., 2017). As for the effect of the energy commodities, the results are qualitatively the same, but the effects are slightly more pronounced. While these results may appear counter-intuitive given that Brazil is more of a non-energy commodities exporter (see Table 2), we offer the following explanation: The expansion of the biofuel industry might have played a key role in intensifying the linkages between agricultural commodities and energy commodities, which has led to a relatively stronger linkages between the volatilities of energy commodities and Brazilian CDS spreads (see, among others, Nazlioglu et al., 2013; Han et al., 2015).

In India, the results from Table 6 reveal some important and distinctive aspects of the effects of both commodity and energy volatilities on the volatility of the CDS spreads. The contemporaneous and lagged effects of commodity and energy volatilities are insignificant in all quantiles. However, the negative and significant coefficients associated to the dummy variable at upper quantiles imply that the decline in the price of energy commodities has decreased the volatility of sovereign risk in India. The latter is one of the major consumers of (energy) commodities. The effect of lagged values of the volatility of the CDS spread is significantly positive and exhibits an upward trend.

The results from Table 7 show that the contemporaneous effect of commodity volatility on the volatility of the Chinese CDS spreads is significantly positive at the 70th and 90th quantiles, whereas past values of commodity volatility have a negative impact at the same upper quantiles. The decline in energy commodity prices has no effect on the volatility of the Chinese CDS spreads at any quantile, suggesting that

investors and policy-makers no need to worry about the effecting of declining energy prices on the volatility of CDS spreads in China. This result seems counter-intuitive given that China represents one of the major oil-consuming countries; however, the contribution of commodity exports as % of merchandise exports is quite marginal (8%). As for the effect of lagged values of the volatility of the CDS spread, it is significantly positive and exhibits stronger relation as we move higher in the quantiles. For the case of the contribution of energy price volatility to the volatility of Chinese sovereign risk, the results show that the contemporaneous and lagged effect of energy volatility are more concentrated at the extreme upper quantile of 0.95.

To summarize,¹⁰ we uncover in all cases a heterogeneous dependence between the volatilities of commodity and energy prices and that of BRIC sovereign CDS spreads; however, for the cases of energy-importing countries (India and China), the dependence is significant only at some upper quantiles, whereas it is significant at both middle and upper quantiles for commodity-exporter (Brazil) and energy-exporter (Russia). In all cases, the energy index exhibits higher impacts on the volatility transmission of CDS spreads than the commodity index (Bouri et al., 2017). The reported heterogeneity in the results suggests the appropriateness of using a quantile based approach. As for the impact of the mid-2014 energy price decline, it is significant only at some high quantiles. Importantly, it was significantly positive in Brazil and Russia, whereas we observed a negative impact in India and no impact in

¹⁰ Unreported results indicate that the Pseudo-R² is higher than 0.79 in all cases and across quantiles, suggesting that the quantile regression explains at least 79% in every quantile. Particularly, for the case of Brazil and Russia, the Pseudo-R² is around 0.84 in lower quantiles and reaches 0.88 in some upper quantiles.

China. These results are not consistent with the overall view that all large energy importers have benefited from the decline in energy prices through the reduction in the volatility of their sovereign risk, given that only India has experienced a decrease in the volatility of its sovereign risk. In contrast, the volatility of sovereign risk in both commodity-exporter and energy exporter has increased in the period that followed the mid-2014 energy price decline. This finding suggests that the decline in energy prices had more effect in commodity/energy exporters.

Compared to prior studies, we added to Sharma and Thuraiamy (2013) by indicating that energy price volatility can be used as a predictor for the volatility of CDS spreads while considering the impact of the mid-2014 energy price decline. We added to the work of Amstad et al. (2016) by focusing on the direct role of volatility of commodity and energy prices, part of global risk factors (Amstad et al., 2016), in explaining the volatility of sovereign CDS spread changes, while differentiating between BRIC energy importers and exporters. In fact, we showed that the sign of this effect differs between commodity export- and commodity import-dependent countries. While Liu et al. (2016) show that oil price volatility can accentuate the volatility of country risk ratings, we revealed that lower energy price had reduced the volatility of sovereign risk of commodity/energy importers. The fact that we accounted for the difference in the impact across lower, middle, and upper quantiles allowed us to extend the findings reported by Bouri et al. (2017).

We assessed the robustness of the above results by using another proxy for volatility. After using the volatility series extracted from the Exponential GARCH model, unreported results revealed that the overall results remains qualitatively the same.

6. Conclusion

This study was motivated by academic and press articles on the role of (energy) commodities shocks, part of global risk factors (Amstad et al., 2016; Global Risks Report, 2016), in explaining sovereign CDS spread changes in countries that are dependent on energy and commodity exportations/importations such as BRIC. We employed a GARCH-quantile approach to examine the overall dependence structure between the volatilities of commodities and sovereign risk, accounting for the impact of the mid-2014 energy price decline. One of the key advantages of this approach is its ability to provide more nuanced modelling of the volatility interdependence than in OLS regression. We therefore extended the work of Bouri et al. (2017) and drew a more nuanced – *beyond the average* – picture about the volatility linkages between commodities price volatility and the volatility of CDS spreads in BRIC countries. Our analysis showed that this unprecedented application of the GARCH-quantile regression to study the volatility interdependencies between commodity/energy markets and CDS spread changes is very useful to capture the difference between upper and lower tail dependence in the volatility transmission, and in uncovering the impact of the mid-2014 energy price decline on the different quantiles.

The empirical analysis reveals several important findings. First, we confirmed the presence of a common component of systematic risk for both oil-exporting and oil-importing large emerging countries (Amstad et al., 2016), which is the volatility of energy (commodities). Second, we also showed the importance of domestic fundamentals, such as the level of commodity/energy dependence (Bouri et al., 2017), in shaping the volatility of sovereign risk in BRIC countries. Third, we indicated that the mid-2014 energy price decline had affected the volatility linkages, and decreased (increased) the volatility of sovereign risk of energy/commodity importers (exporters).

The above findings are not only useful for policy directed research, but also for traders and investors who can build upon our nuanced results to construct hedging based strategy in relevant quantiles. For policymakers, uncovering the volatility relationship between commodities and sovereign CDS markets is useful for their design of suitable

regulatory structures. Given the reported evidence on the increase in the volatility linkages in some cases (Brazil and Russia) following the considerable decrease in energy prices in mid-2014, it emerges the crucial need for policy-makers in those countries to devote more of their regulatory focus on the commodity and energy markets in order to cushion adverse spillover effects. It is highly recommended that energy-dependent countries such as Russia should diversify their economy away from energy and gas, toward non-energy sectors by promoting, among others, tourism, tech-related and heavy machinery exportation. The high dependence on fuel sector cannot be sustained; ultimately, the fall of energy prices will eventually affect the inflow of capitals, causing a deficit in the balance of payments, which would lead to a deteriorating in the value of the currency. Such a currency exchange risk would undermine the demand on Russian assets. Investors and traders who can be exposed to severe economic losses from risk events in emerging economies can more effectively re-weight their international portfolio now they better understand the effect of specific global risk factors (i.e. the volatility of energy commodities) that affect the volatility of BRIC sovereign CDS spreads.

While we documented a significant effect of volatility from commodities and energy commodities to the volatility of BRIC sovereign risk in the quantiles, we couldn't rule out the possibility that our findings might be affected by other economic, financial, and political factors. Importantly, our findings would open the door for new in-depth research on the effect of lower commodity/energy prices on sovereign risk of small emerging countries.

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